MAHATMA GANDHI UNIVERSITY, KOTTAYAM

QP CORD 25900153 MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCPES100 - PHYSICAL EDUCATION-FOUNDATION & CAREER PROSPECT

(2024 ADMISSION ONWARDS)

Part A **Duration: 1 Hour**

Multiple Choice Questions

Answer All Questions

Each Question carries 1 mark

1. Regular participation in physical education can help prevent which chronic disease?
   (a) Diabetes
2. The term "muscular endurance" refers to.....
   b). The ability of a muscle to sustain repeated contractions over time
3. The ability to perform a movement with accuracy and smoothness is called ……
   a) Flexibility
4. What is one major benefit of maintaining a healthy lifestyle?
   a) Better overall physical and mental well-being
5. The spectrum of health helps in understanding
   c) The progression from good health to illness and death
6. Why is regular physical activity crucial in managing hypertension?
   b) It improves blood vessel health, reduces stress, and helps maintain a healthy weight
7. The 600-yard run/walk test in the AAHPERD battery is used to assess
   b) Cardiovascular endurance
8. Which body type has the highest tendency to store fat?
   c) Endomorph
9. A child falls from a tree and their leg appears bent at an unnatural angle. What should you do?
   b) Immobilize the limb and seek medical help
10. High body fat percentage increases the risk of which of the following?
    d) All of the above **[10x1 = 10]**

Part B

Short Answer Type Questions

Answer any 5 Questions

Each Question carries 2 marks

11. What type of exercise is best for improving endurance?

    The best type of exercise for improving endurance is aerobic exercise.

    Examples include: – Running – Swimming – Cycling – Brisk walking– Rowing

12. What is anaerobic exercise?

    Anaerobic exercise is a type of high-intensity physical activity where the body's demand for oxygen exceeds the oxygen supply available, so the body relies on energy sources stored in the muscles rather than oxygen from breathing. Examples include: – Sprinting– Weightlifting– High-intensity interval training (HIIT)– Jumping or plyometrics. It is mainly used to build strength, power, and muscle mass, and to improve short bursts of speed.

13. What is the definition of health according to the World Health Organization?

    According to the World Health Organization (WHO), health is defined as: "A state of complete physical, mental and social well-being and not merely the absence of disease or infirmity."

14. What is adaptation in the context of physical training?

Adaptation in the context of physical training refers to the body's physiological response to repeated exercise, where it gradually becomes more efficient and better able to handle physical stress. For example:– Muscles become stronger– Endurance improves– The cardiovascular system works more efficiently. This process allows an individual to perform exercises with greater ease or at higher intensities over time.

15. What is the circadian rhythm?

Circadian rhythm is the natural, internal process that regulates the sleep-wake cycle and repeats roughly every 24 hours. It influences various bodily functions such as:– Sleep patterns– Hormone release– Body temperature– Digestion. The circadian rhythm is mainly controlled by the brain's hypothalamus, and it is affected by external cues like light and darkness.

16. Identify two postural deformities in children and suggest ways to prevent them.

Two common postural deformities in children are:

1. **Scoliosis** (Curvature of the spine)
    - **Prevention**:
      – Encourage good posture habits (sitting and standing with a straight back).
      – Limit prolonged sitting in one position (e.g., for long periods of screen time).
      – Promote activities that strengthen the core and back muscles (e.g., swimming, yoga).
2. **Kyphosis** (Excessive forward curvature of the upper spine, often called "hunchback")
    - **Prevention**:
      – Ensure children sit with their backs straight and shoulders back.
      – Encourage activities that stretch and strengthen the upper back, such as exercises for posture improvement.
      – Avoid heavy backpacks or improper carrying techniques, which can strain the back.

17. What is correct posture, and why is it important?

Correct posture refers to the alignment of the body in such a way that the bones, muscles, and joints are positioned to avoid unnecessary strain. In a standing position, this means:– The head is aligned with the spine. – Shoulders are relaxed and level. – The back is straight with a natural curve in the lower back. – Knees are slightly bent, and weight is evenly distributed across both feet.

Importance of Correct Posture: Prevents musculoskeletal issues, improves breathing and digestion, Boosts energy levels, Enhances confidence.

[5x2 = 10]

Part C
Short Essay Type Questions
Answer any 3 Questions
Each Question carries 5 marks.

18 How does Physical Education support sustainable health habits in the face of rising obesity and mental health issues?

Physical Education plays a critical role in supporting sustainable health habits, especially in addressing the rising concerns of obesity and mental health issues. Here's how it contributes:

1. Promoting Regular Physical Activity 2. Education on Healthy Lifestyle Choices 3. Mental Health Benefits, 4. Building Healthy Habits Early, 5. Stress Relief and Cognitive Function,

19. What is nutrition? Explain the classification of nutrients and their role in maintaining health.

Nutrition refers to the process by which the body takes in and uses food for growth, energy, and overall health. It involves the intake of essential substances from food, which are then absorbed and utilized by the body for various physiological functions.

Classification of Nutrients: Nutrients are substances required by the body for growth, energy, and proper functioning. They are classified into macronutrients and micronutrients, each playing a crucial role in maintaining health.

Explain its role and sources of macronutrients and micronutrients

20. What are dietary supplements? Discuss their potential benefits and risks?

Dietary supplements are products taken orally that contain nutrients or other substances intended to supplement the diet. They can include vitamins, minerals, herbs, amino acids, enzymes, or other bioactive compounds. They come in various forms such as tablets, capsules, powders, liquids, and gummies.

Examples include: – Vitamin D tablets – Iron supplements – Protein powders – Omega-3 capsules – Herbal supplements like echinacea or ginseng

Potential Benefits of Dietary Supplements

1. Correct Nutrient Deficiencies
2. Support Specific Health Conditions
3. Enhance Athletic Performance
4. Boost Immune Function and General Wellness

Potential Risks of Dietary Supplements

1. Overconsumption or Toxicity
2. Interactions with Medications
3. False Health Claims and Lack of Regulation
4. Delay in Seeking Medical Treatment

21. How does scoliosis impact muscle balance?

Scoliosis is a condition characterized by an abnormal sideways curvature of the spine, often in an "S" or "C" shape. This spinal misalignment can have a significant impact on muscle balance in the body. Here's how:

1. Uneven Muscle Tension,
2. Compensatory Muscle Imbalances
3. Reduced Range of Motion and Function
4. Increased Risk of Fatigue and Pain

22. Analyze the effectiveness of the Harvard Step Test in assessing cardiovascular endurance.

The Harvard Step Test is a simple, submaximal exercise test used to measure cardiovascular endurance (the ability of the heart and lungs to supply oxygen during sustained physical activity).

Procedure:
– The participant steps up and down on a 45 cm (18 inch) bench at a rate of 30 steps per minute for 5 minutes (or until exhaustion).
– After exercise, the heart rate recovery is measured at set intervals, and the results are used to assess cardiovascular fitness. **[3x5 =15]**

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